DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO.3

Case study: Airline Management System

Mini Project: Centralized Club Management System

Name: Nishtha Pramod Parve

PRN: 124B1F003

Case Study: Airline Management System

Aim: SQL Queries- All types of Join, Sub-Query and View:
a. Write at least 10 SQL queries for suitable database application using SQL DML statements using joins and subqueries.
b. Write queries using group by and having clause.

1. SQL Joins

A **JOIN** is used to combine data from multiple tables based on a related column.

Types of Joins:

- **INNER JOIN**: Returns matching records from both tables
- **LEFT JOIN**: Returns all records from left table + matched from right
- **RIGHT JOIN**: Returns all records from right table + matched from left
- **FULL JOIN**: Returns all records when there is a match in either table

Joins are useful when data is stored in multiple tables.

2. Subqueries

A **subquery** is a query inside another query.

- Used in SELECT, WHERE, FROM
- Helps to filter or calculate dynamic values

Example: Find students with marks above average.

3. GROUP BY Clause

- Used to group rows with same values
- Mostly used with aggregate functions like:
    - SUM(), AVG(), COUNT(), MAX(), MIN()

4. HAVING Clause

- Used with GROUP BY
- Filters grouped data (like WHERE but after grouping)

5. Views

A **VIEW** is a virtual table created from a query.

- Does not store data physically
- Used for security and simplification

```
mysql> USE airline_management;
Database changed
mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_airline_management |
+-----------------------------+
| airline                     |
| airline_copy                |
| booking                     |
| flight                      |
| passenger                   |
+-----------------------------+
5 rows in set (0.04 sec)

mysql> select * from airline;
+------------+--------------+---------+--------------+
| airline_id | airline_name | country | founded_year |
+------------+--------------+---------+--------------+
|          1 | Air India    | India   |         NULL |
|          2 | Indigo       | India   |         NULL |
|          3 | Spicejet     | India   |         NULL |
|          4 | Airindia     | India   |         NULL |
+------------+--------------+---------+--------------+
4 rows in set (0.02 sec)
```

```
mysql> select * from booking;
+------------+-----------+--------------+-------------+---------+--------------+
| booking_id | flight_id | passenger_id | booking_date | seat_no | travel_class |
+------------+-----------+--------------+-------------+---------+--------------+
|       1001 |       101 |          501 | 2024-02-01  | 12A     | Economy      |
|       1002 |       105 |          789 | 2023-09-08  | 14B     | Economy      |
|       1003 |       123 |          890 | 2025-07-07  | 13A     | Economy      |
+------------+-----------+--------------+-------------+---------+--------------+
3 rows in set (0.01 sec)

mysql> select * from flight;
+-----------+------------+---------+-------------+-------------+-------------+-------------+-----------+
| flight_id | airline_id | source  | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+---------+-------------+-------------+-------------+-------------+-----------+
|       101 |          1 | Mumbai  | Lucknow     | 10:00:00    | 12:00:00    | Domestic    |       180 | Scheduled |
|       102 |          2 | kolkata | nagpur      | 12:00:00    | 04:00:00    | Domestic    |       180 | Scheduled |
```

```
|      104 |         4 | mumbai  | chennai     |
06:00:00       | 09:00:00     | Domestic    |
180 | Scheduled |

+-----------+------------+---------+-------------+---
-------------+--------------+-------------+-----------
--+-----------+

3 rows in set (0.00 sec)


mysql> select * from passenger;

+--------------+----------------+--------+------+----
---------+-------------+

| passenger_id | passenger_name |
gender | age  | passport_no | nationality
|
```

```
+--------------+----------------+--------+------+----
---------+-------------+

|        501 | Rahul Mehta    | M      |   35 |
P1234567    | Indian      |

|        678 | Rohit          | M      |   23 |
P85666      | Indian      |

|        888 | Jiya           | F      |   46 |
P45677      | Indian      |

+--------------+----------------+--------+------+----
---------+-------------+

3 rows in set (0.00 sec)
```

SQL Queries (Joins + Subqueries)

1. INNER JOIN (Passenger + Booking)

```
SELECT p.passenger_name, b.booking_id, b.seat_no

FROM passenger p

INNER JOIN booking b

ON p.passenger_id = b.passenger_id;
```

```
+----------------+------------+---------+
| passenger_name | booking_id | seat_no |
+----------------+------------+---------+
| Rahul Mehta    |       1001 | 12A     |
+----------------+------------+---------+
1 row in set (0.00 sec)
```

2. JOIN (Booking + Flight)

SELECT b.booking_id, f.source, f.destination

FROM booking b

JOIN flight f

ON b.flight_id = f.flight_id;

```
+------------+--------+-------------+
| booking_id | source | destination |
+------------+--------+-------------+
|       1001 | Mumbai | Lucknow     |
+------------+--------+-------------+
1 row in set (0.00 sec)
```

3. JOIN (Flight + Airline)

SELECT f.flight_id, a.airline_name, f.source, f.destination

FROM flight f

JOIN airline a

ON f.airline_id = a.airline_id;

```
+-----------+--------------+---------+-------------+
| flight_id | airline_name | source  | destination |
+-----------+--------------+---------+-------------+
|       101 | Air India    | Mumbai  | Lucknow     |
|       102 | Indigo       | kolkata | nagpur      |
|       104 | Airindia     | mumbai  | chennai     |
+-----------+--------------+---------+-------------+
3 rows in set (0.00 sec)
```

4. MULTIPLE JOIN (Passenger + Booking + Flight)

SELECT p.passenger_name, f.source, f.destination, b.seat_no

FROM passenger p

JOIN booking b ON p.passenger_id = b.passenger_id

JOIN flight f ON b.flight_id = f.flight_id;

```
+----------------+--------+-------------+---------+
| passenger_name | source | destination | seat_no |
+----------------+--------+-------------+---------+
| Rahul Mehta    | Mumbai | Lucknow     | 12A     |
+----------------+--------+-------------+---------+
1 row in set (0.01 sec)
```

5. LEFT JOIN (Passengers without booking)

```
SELECT p.passenger_name, b.booking_id

FROM passenger p

LEFT JOIN booking b

ON p.passenger_id = b.passenger_id;
```

passenger_name	booking_id
Rahul Mehta	1001
Rohit	NULL
Jiya	NULL

3 rows in set (0.00 sec)

6. RIGHT JOIN (Flights with bookings)

```
SELECT f.flight_id, b.booking_id

FROM flight f

RIGHT JOIN booking b

ON f.flight_id = b.flight_id;
```

flight_id	booking_id
101	1001
NULL	1002
NULL	1003

3 rows in set (0.00 sec)

7. Subquery (Passengers who have bookings)

```
SELECT passenger_name

FROM passenger

WHERE passenger_id IN (

   SELECT passenger_id FROM booking

);
```

```
+----------------+
| passenger_name |
+----------------+
| Rahul Mehta    |
+----------------+
1 row in set (0.00 sec)
```

8. Subquery (Flights that are booked)

SELECT flight_id, source, destination

FROM flight

WHERE flight_id IN (

  SELECT flight_id FROM booking

);

```
+-----------+--------+-------------+
| flight_id | source | destination |
+-----------+--------+-------------+
|       101 | Mumbai | Lucknow     |
+-----------+--------+-------------+
1 row in set (0.00 sec)
```

9. Subquery (Passengers NOT having bookings)

SELECT passenger_name

FROM passenger

WHERE passenger_id NOT IN (

  SELECT passenger_id FROM booking

);

```
+----------------+
| passenger_name |
+----------------+
| Rohit          |
| Jiya           |
+----------------+
2 rows in set (0.00 sec)
```

10. Subquery with MAX (Latest Booking)

SELECT *

FROM booking

```
WHERE booking_date = (
   SELECT MAX(booking_date) FROM booking
);
```

booking_id	flight_id	passenger_id	booking_date	seat_no	travel_class
1003	123	890	2025-07-07	13A	Economy

1 row in set (0.02 sec)

11. Subquery with EXISTS

```
SELECT passenger_name
FROM passenger p
WHERE EXISTS (
   SELECT 1
   FROM booking b
   WHERE p.passenger_id = b.passenger_id
);
```

passenger_name
Rahul Mehta

1 row in set (0.00 sec)

12. JOIN + Subquery (Flights of specific airline)

```
SELECT flight_id, source, destination
FROM flight
WHERE airline_id = (
   SELECT airline_id
   FROM airline
   WHERE airline_name = 'Air India'
);
```

```
+-----------+--------+-------------+
| flight_id | source | destination |
+-----------+--------+-------------+
|       101 | Mumbai | Lucknow     |
+-----------+--------+-------------+
1 row in set (0.01 sec)
```

SQL Queries using GROUP BY and HAVING

1. Count bookings per passenger

SELECT passenger_id, COUNT(booking_id) AS total_bookings

FROM booking

GROUP BY passenger_id;

```
+--------------+----------------+
| passenger_id | total_bookings |
+--------------+----------------+
|          501 |              1 |
|          789 |              1 |
|          890 |              1 |
+--------------+----------------+
3 rows in set (0.03 sec)
```

2. Passengers with more than 1 booking (HAVING)

SELECT passenger_id, COUNT(booking_id) AS total_bookings

FROM booking

GROUP BY passenger_id

HAVING COUNT(booking_id) > 1;

```
mysql> SELECT passenger_id, COUNT(booking_id) AS total_bookings
    -> FROM booking
    -> GROUP BY passenger_id
    -> HAVING COUNT(booking_id) > 1;
Empty set (0.01 sec)
```

3. Count bookings per flight

SELECT flight_id, COUNT(booking_id) AS total_bookings

FROM booking

GROUP BY flight_id;

```
+-----------+----------------+
| flight_id | total_bookings |
+-----------+----------------+
|       101 |              1 |
|       105 |              1 |
|       123 |              1 |
+-----------+----------------+
3 rows in set (0.02 sec)
```

4. Flights with more than 1 booking (HAVING)

SELECT flight_id, COUNT(booking_id) AS total_bookings

FROM booking

GROUP BY flight_id

HAVING COUNT(booking_id) > 1;

```
mysql> SELECT flight_id, COUNT(booking_id) AS total_bookings
    -> FROM booking
    -> GROUP BY flight_id
    -> HAVING COUNT(booking_id) > 1;
Empty set (0.01 sec)
```

5. Count passengers by gender

SELECT gender, COUNT(passenger_id) AS total_passengers

FROM passenger

GROUP BY gender;

```
+--------+------------------+
| gender | total_passengers |
+--------+------------------+
| M      |                2 |
| F      |                1 |
+--------+------------------+
2 rows in set (0.00 sec)
```

6. Average age of passengers by gender

SELECT gender, AVG(age) AS avg_age

FROM passenger

GROUP BY gender;

```
+--------+---------+
| gender | avg_age |
+--------+---------+
| M      | 29.0000 |
| F      | 46.0000 |
+--------+---------+
2 rows in set (0.01 sec)
```

7. Airlines with number of flights

SELECT airline_id, COUNT(flight_id) AS total_flights

FROM flight

GROUP BY airline_id;

```
+------------+---------------+
| airline_id | total_flights |
+------------+---------------+
|          1 |             1 |
|          2 |             1 |
|          4 |             1 |
+------------+---------------+
3 rows in set (0.00 sec)
```

8. Airlines having more than 1 flight (HAVING)

SELECT airline_id, COUNT(flight_id) AS total_flights

FROM flight

GROUP BY airline_id

HAVING COUNT(flight_id) > 1;

```
mysql> SELECT airline_id, COUNT(flight_id) AS total_flights
    -> FROM flight
    -> GROUP BY airline_id
    -> HAVING COUNT(flight_id) > 1;
Empty set (0.00 sec)
```

9. Number of bookings by travel class

SELECT travel_class, COUNT(booking_id) AS total_bookings

FROM booking

GROUP BY travel_class;

```
+--------------+----------------+
| travel_class | total_bookings |
+--------------+----------------+
| Economy      |              3 |
+--------------+----------------+
1 row in set (0.00 sec)
```

10. Travel class with more than 1 booking (HAVING)

SELECT travel_class, COUNT(booking_id) AS total_bookings

FROM booking

GROUP BY travel_class

HAVING COUNT(booking_id) > 1;

```
+--------------+----------------+
| travel_class | total_bookings |
+--------------+----------------+
| Economy      |              3 |
+--------------+----------------+
1 row in set (0.00 sec)
```

11. Join + GROUP BY (Passengers per flight)

SELECT f.flight_id, COUNT(b.passenger_id) AS total_passengers

FROM flight f

JOIN booking b ON f.flight_id = b.flight_id

GROUP BY f.flight_id;

```
+-----------+------------------+
| flight_id | total_passengers |
+-----------+------------------+
|       101 |                1 |
+-----------+------------------+
1 row in set (0.00 sec)
```

12. Join + GROUP BY + HAVING

SELECT f.flight_id, COUNT(b.passenger_id) AS total_passengers

FROM flight f

JOIN booking b ON f.flight_id = b.flight_id

GROUP BY f.flight_id

HAVING COUNT(b.passenger_id) > 1;

```
mysql> SELECT f.flight_id, COUNT(b.passenger_id) AS total_passengers
    -> FROM flight f
    -> JOIN booking b ON f.flight_id = b.flight_id
    -> GROUP BY f.flight_id
    -> HAVING COUNT(b.passenger_id) > 1;
Empty set (0.00 sec)
```

Mini Project: Centralized Club Management System

```
mysql> USE club_management;

Database changed

mysql> SHOW TABLES;

+--------------------------+

| Tables_in_club_management |

+--------------------------+

| attendance               |

| budget_request           |

| club                     |

| club_coordinator         |

| club_lead                |

| club_members             |

| event                    |

| reports                  |

| super_admin              |

+--------------------------+

9 rows in set (0.07 sec)


mysql> select * from attendance;

+---------------+-----------+----------+---------+

| attendance_id | member_id | event_id | status  |

+---------------+-----------+----------+---------+

|             2 |         2 |        1 | Absent  |

|             3 |         3 |        2 | Present |

|             4 |         4 |        3 | Present |

|             5 |         5 |        4 | Absent  |

+---------------+-----------+----------+---------+

4 rows in set (0.00 sec)


mysql> select * from budget_request;

+-----------+----------+----------------+---------+--
-------+------------+-----------------+

| budget_id | event_id | coordinator_id |
club_id | lead_id | submission |
approved_status |

+-----------+----------+----------------+---------+--
-------+------------+-----------------+

|         1 |        1 |            101 |       1 |     201
| 2026-04-01 | Pending        |

|         2 |        2 |            102 |       2 |     202
| 2026-04-05 | Approved       |

|         3 |        3 |            103 |       3 |     203
| 2026-04-07 | Pending        |

|         4 |        4 |            104 |       4 |     204
| 2026-04-10 | Approved       |
```

```
|        5 |        5 |           105 |       5 |     205
| 2026-04-12 | Pending        |

+-----------+----------+----------------+---------+--
-------+------------+-----------------+

5 rows in set (0.02 sec)


mysql> select * from club;

+---------+---------------+---------+----------------
+
| club_id | club_name     | lead_id |
coordinator_id |

+---------+---------------+---------+----------------
+

|       1 | Tech Club     |     201 |           101 |

|       2 | Cultural Club |     202 |           102
|

|       3 | Sports Club   |     203 |           103
|

|       4 | Music Club    |     204 |           104
|

|       5 | Art Club      |     205 |           105 |

+---------+---------------+---------+----------------
+

5 rows in set (0.01 sec)

mysql> select * from club_coordinator;

+----------------+-----------------+----------------
+----------+---------+

| coordinator_id | coordinator_name |
email          | password | club_id |

+----------------+-----------------+----------------
+----------+---------+

|           101 | Anita           |
anita@gmail.com | pass     |       1 |

|           102 | Priya           |
priya@gmail.com | pass     |       2 |

|           103 | Rohit           |
rohit@gmail.com | pass     |       3 |

|           104 | Sneha           |
sneha@gmail.com | pass     |       4 |

|           105 | Vikas           |
vikas@gmail.com | pass     |       5 |

+----------------+-----------------+----------------
+----------+---------+

5 rows in set (0.01 sec)


mysql> select * from club_lead;

+---------+-----------+-----------------+----------
+---------+----------------+

| lead_id | lead_name | email           |
password | club_id | coordinator_id |

+---------+-----------+-----------------+----------
+---------+----------------+

|     201 | Rahul     | rahul@gmail.com  |
pass     |       1 |           101 |

|     202 | Karan     | karan@gmail.com  |
pass     |       2 |           102 |

|     203 | Amit      | amit@gmail.com   |
pass     |       3 |           103 |

|     204 | Pooja     | pooja@gmail.com  |
pass     |       4 |           104 |

|     205 | Neeraj    | neeraj@gmail.com |
pass     |       5 |           105 |

+---------+-----------+-----------------+----------
+---------+----------------+
```

```
5 rows in set (0.01 sec)


mysql> select * from club_members;
+------+--------+----------------+------------+-----
--------+----------+---------+
| m_id | m_name | m_email        |
m_password | event_roll  | domain   |
club_id |
+------+--------+----------------+------------+-----
--------+----------+---------+
|    1 | Aman   | aman@gmail.com  | 123
| Participant | AI      |       1 |
|    2 | Riya   | riya@gmail.com  | 123
| Volunteer   | Design   |       1 |
|    3 | Neha   | neha@gmail.com  | 123
| Participant | Dance    |       2 |
|    4 | Arjun  | arjun@gmail.com | 123
| Player      | Cricket  |       3 |
|    5 | Meera  | meera@gmail.com | 123
| Singer      | Vocal    |       4 |
|    6 | Kabir  | kabir@gmail.com | 123
| Artist      | Painting |       5 |
+------+--------+----------------+------------+-----
--------+----------+---------+
6 rows in set (0.00 sec)


mysql> select * from event;
+----------+----------------+------------+------------
+-----------+---------+---------+-----------+
| event_id | event_name     | start_date |
end_date   | report_id | lead_id | club_id
| budget_id |
+----------+----------------+------------+------------
+-----------+---------+---------+-----------+
|        1 | Hackathon      | 2026-04-01 |
2026-04-02 |      NULL |     201 |       1 |
NULL |
|        2 | Dance Fest     | 2026-04-05 |
2026-04-06 |      NULL |     202 |       2 |
NULL |
|        3 | Football Match | 2026-04-07 |
2026-04-08 |      NULL |     203 |       3 |
NULL |
|        4 | Music Night    | 2026-04-10 |
2026-04-11 |      NULL |     204 |       4 |
NULL |
|        5 | Art Exhibition | 2026-04-12 |
2026-04-13 |      NULL |     205 |       5 |
NULL |
+----------+----------------+------------+------------
+-----------+---------+---------+-----------+
5 rows in set (0.00 sec)


mysql> select * from reports;
+-----------+----------+---------+----------------+
| report_id | event_id | lead_id |
coordinator_id |
+-----------+----------+---------+----------------+
|         1 |        1 |     201 |            101 |
|         2 |        2 |     202 |            102 |
|         3 |        3 |     203 |            103 |
|         4 |        4 |     204 |            104 |
|         5 |        5 |     205 |            105 |
+-----------+----------+---------+----------------+
```

```
5 rows in set (0.00 sec)


mysql> select * from super_admin;

+----------+------------+-----------------+----------
+----------------+

| admin_id | admin_name | email           |
password | overall_budget |

+----------+------------+-----------------+----------
+----------------+

|        1 | Admin1     | admin1@gmail.com
| pass123  |      100000.00 |
```

```
|        2 | Admin2     | admin2@gmail.com
| pass123  |      150000.00 |

|        3 | Admin3     | admin3@gmail.com
| pass123  |      200000.00 |

|        4 | Admin4     | admin4@gmail.com
| pass123  |      120000.00 |

|        5 | Admin5     | admin5@gmail.com
| pass123  |      180000.00 |

+----------+------------+-----------------+----------
+----------------+

5 rows in set (0.00 sec)
```

SQL Queries (JOIN + SUBQUERY)

1. INNER JOIN (Club + Lead)

```
mysql> SELECT c.club_name, l.lead_name
    -> FROM Club c
    -> INNER JOIN Club_Lead l ON c.lead_id = l.lead_id;
+---------------+-----------+
| club_name     | lead_name |
+---------------+-----------+
| Tech Club     | Rahul     |
| Cultural Club | Karan     |
| Sports Club   | Amit      |
| Music Club    | Pooja     |
| Art Club      | Neeraj    |
+---------------+-----------+
5 rows in set (0.00 sec)
```

2. INNER JOIN (Club + Coordinator)

```
mysql> SELECT c.club_name, co.coordinator_name
    -> FROM Club c
    -> INNER JOIN Club_Coordinator co ON c.coordinator_id = co.coordinator_id;
+---------------+------------------+
| club_name     | coordinator_name |
+---------------+------------------+
| Tech Club     | Anita            |
| Cultural Club | Priya            |
| Sports Club   | Rohit            |
| Music Club    | Sneha            |
| Art Club      | Vikas            |
+---------------+------------------+
5 rows in set (0.00 sec)
```

3. JOIN (Event + Club)

```
mysql> SELECT e.event_name, c.club_name
    -> FROM Event e
    -> JOIN Club c ON e.club_id = c.club_id;
+----------------+---------------+
| event_name     | club_name     |
+----------------+---------------+
| Hackathon      | Tech Club     |
| Dance Fest     | Cultural Club |
| Football Match | Sports Club   |
| Music Night    | Music Club    |
| Art Exhibition | Art Club      |
+----------------+---------------+
5 rows in set (0.00 sec)
```

4. JOIN (Attendance + Members + Event)

```
mysql> SELECT m.m_name, e.event_name, a.status
    -> FROM Attendance a
    -> JOIN Club_members m ON a.member_id = m.m_id
    -> JOIN Event e ON a.event_id = e.event_id;
+--------+----------------+---------+
| m_name | event_name     | status  |
+--------+----------------+---------+
| Riya   | Hackathon      | Absent  |
| Neha   | Dance Fest     | Present |
| Arjun  | Football Match | Present |
| Meera  | Music Night    | Absent  |
+--------+----------------+---------+
4 rows in set (0.00 sec)
```

5. LEFT JOIN (All clubs even if no event)

```
mysql> SELECT c.club_name, e.event_name
    -> FROM Club c
    -> LEFT JOIN Event e ON c.club_id = e.club_id;
+---------------+----------------+
| club_name     | event_name     |
+---------------+----------------+
| Tech Club     | Hackathon      |
| Cultural Club | Dance Fest     |
| Sports Club   | Football Match |
| Music Club    | Music Night    |
| Art Club      | Art Exhibition |
+---------------+----------------+
5 rows in set (0.00 sec)
```

6. RIGHT JOIN (All events with club info)

```
mysql> SELECT e.event_name, c.club_name
    -> FROM Event e
    -> RIGHT JOIN Club c ON e.club_id = c.club_id;
+----------------+---------------+
| event_name     | club_name     |
+----------------+---------------+
| Hackathon      | Tech Club     |
| Dance Fest     | Cultural Club |
| Football Match | Sports Club   |
| Music Night    | Music Club    |
| Art Exhibition | Art Club      |
+----------------+---------------+
5 rows in set (0.00 sec)
```

7. SUBQUERY (Members in Tech Club)

```
mysql> SELECT m_name
    -> FROM Club_members
    -> WHERE club_id = (
    ->     SELECT club_id FROM Club WHERE club_name = 'Tech Club'
    -> );
+--------+
| m_name |
+--------+
| Aman   |
| Riya   |
+--------+
2 rows in set (0.01 sec)
```

8. SUBQUERY (Events handled by Rahul)

```
mysql> SELECT event_name
    -> FROM Event
    -> WHERE lead_id = (
    ->     SELECT lead_id FROM Club_Lead WHERE lead_name = 'Rahul'
    -> );
+------------+
| event_name |
+------------+
| Hackathon  |
+------------+
1 row in set (0.00 sec)
```

9. SUBQUERY (Approved budget events)

```
mysql> SELECT event_name
    -> FROM Event
    -> WHERE event_id IN (
    ->     SELECT event_id FROM Budget_Request WHERE approved_status = 'Approved'
    -> );
+-------------+
| event_name  |
+-------------+
| Dance Fest  |
| Music Night |
+-------------+
2 rows in set (0.01 sec)
```

10. JOIN + CONDITION

```
mysql> SELECT m.m_name, c.club_name
    -> FROM Club_members m
    -> JOIN Club c ON m.club_id = c.club_id
    -> WHERE c.club_name = 'Tech Club';
+--------+-----------+
| m_name | club_name |
+--------+-----------+
| Aman   | Tech Club |
| Riya   | Tech Club |
+--------+-----------+
2 rows in set (0.00 sec)
```

11. MULTI JOIN (Club + Lead + Coordinator)

```
mysql> SELECT c.club_name, l.lead_name, co.coordinator_name
    -> FROM Club c
    -> JOIN Club_Lead l ON c.lead_id = l.lead_id
    -> JOIN Club_Coordinator co ON c.coordinator_id = co.coordinator_id;
+---------------+-----------+------------------+
| club_name     | lead_name | coordinator_name |
+---------------+-----------+------------------+
| Tech Club     | Rahul     | Anita            |
| Cultural Club | Karan     | Priya            |
| Sports Club   | Amit      | Rohit            |
| Music Club    | Pooja     | Sneha            |
| Art Club      | Neeraj    | Vikas            |
+---------------+-----------+------------------+
5 rows in set (0.00 sec)
```

GROUP BY + HAVING

1 . Count members in each club

```
mysql> SELECT club_id, COUNT(*) AS total_members
    -> FROM Club_members
    -> GROUP BY club_id;
+---------+---------------+
| club_id | total_members |
+---------+---------------+
|       1 |             2 |
|       2 |             1 |
|       3 |             1 |
|       4 |             1 |
|       5 |             1 |
+---------+---------------+
5 rows in set (0.02 sec)
```

2. Clubs having more than 1 member

```
mysql> SELECT club_id, COUNT(*) AS total_members
    -> FROM Club_members
    -> GROUP BY club_id
    -> HAVING COUNT(*) > 1;
+---------+---------------+
| club_id | total_members |
+---------+---------------+
|       1 |             2 |
+---------+---------------+
1 row in set (0.01 sec)
```

3. Count events per club

```
mysql> SELECT club_id, COUNT(*) AS total_events
    -> FROM Event
    -> GROUP BY club_id;
+---------+--------------+
| club_id | total_events |
+---------+--------------+
|       1 |            1 |
|       2 |            1 |
|       3 |            1 |
|       4 |            1 |
|       5 |            1 |
+---------+--------------+
5 rows in set (0.00 sec)
```

4. Clubs with more than 1 event

```
mysql> SELECT club_id, COUNT(*) AS total_events
    -> FROM Event
    -> GROUP BY club_id
    -> HAVING COUNT(*) > 1;
Empty set (0.00 sec)
```

Conclusion

In this assignment, we successfully explored advanced SQL concepts including Joins, Subqueries, Views, Group By, and Having clauses using both the Airline Management System and the Centralized Club Management System. We implemented different types of joins such as INNER JOIN, LEFT JOIN, and RIGHT JOIN to combine data from multiple related tables effectively.

We also used subqueries to perform nested queries for dynamic data retrieval and filtering. The use of GROUP BY and HAVING clauses helped in organizing data into meaningful groups and applying conditions on aggregated results. Additionally, we learned how to create and use views, which act as virtual tables to simplify complex queries and enhance data security.

Overall, this assignment strengthened our understanding of relational database operations and improved our ability to write efficient and structured SQL queries for real-world applications.